PRODUCT CATALOGUE · 2026

HomeMaster

Eleven DIN-rail modules that put Home Assistant inside the electrical cabinet.

ESP32 · ESPHome · Modbus RTU
local logic · open hardware

Designed and manufactured in Ploiești, Romania



Controller plus modules

One controller runs an RS-485 trunk of specialised I/O modules.

Logic lives in the module

Lighting and leak protection keep running when the server is down.

Nothing is locked

Open schematics and firmware. Plain Modbus RTU, any master.

Configured in a browser

USB-C and Chromium. No app, no driver, no licence.

CE marked · declaration of conformity per module · EMC at Idvorsky Laboratories, Belgrade · EN 62368-1 · controllers RED via Kiwa NB 0063 and EN 18031-1:2024

home-master.eu

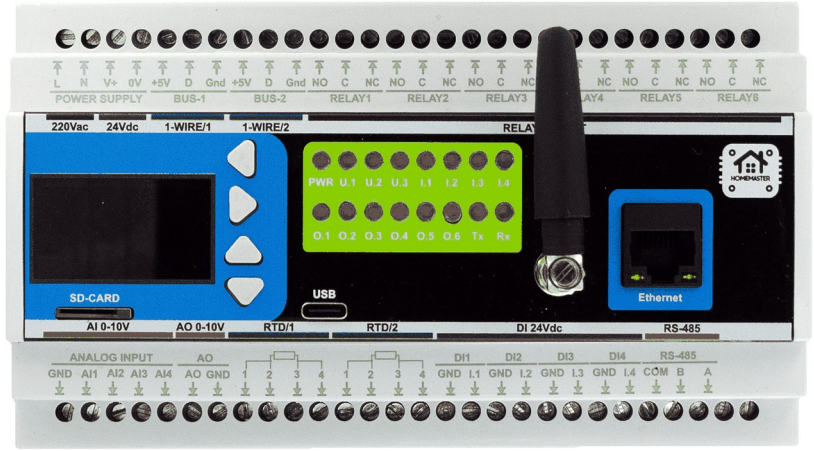
CONTROLLER

MiniPLC

ESP32 controller with onboard I/O, Ethernet and Modbus RTU master

PURPOSE

The flagship controller. Runs the RS-485 trunk for the module line and carries enough onboard I/O to handle a small installation on its own. ESPHome pre-installed; appears in Home Assistant over the native API with no MQTT broker and no manual register mapping.



SPECIFICATIONS

Relays	6 × SPDT, 3 A @ 250 VAC
Digital inputs	4 × 24 V, IEC 61131-2 (ISO1212)
Analogue inputs	4 × 0–10 V, 16-bit (ADS1115)
Analogue output	1 × 0–10 V, 12-bit
RTD	2 × PT100 / PT1000, 2/3/4-wire
1-Wire	2 buses
Network	Ethernet + Wi-Fi
Extras	microSD, OLED, RTC
Power	24 V DC nominal or 85–265 V AC
Bus	RS-485 Modbus RTU master

WHAT MAKES IT DIFFERENT

The controller thinks, the server only shows

Critical logic runs on the MiniPLC. Home Assistant provides dashboards and history — when it is offline or updating, the installation does not stop.

RTD plus analogue I/O on the controller

Uncommon in this class. Temperature sensing and 0–10 V actuation without stacking extra boards.

Improv onboarding

Wi-Fi provisioned over Bluetooth or USB from a browser. No app to install on site.

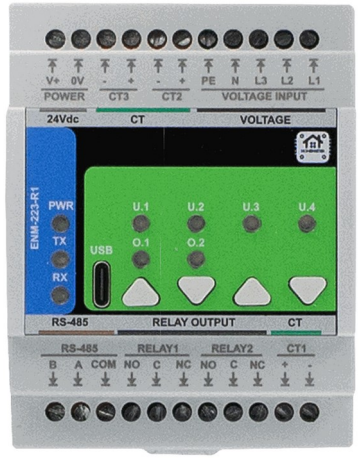
ENERGY

ENM-223-R1

Three-phase energy meter with power quality and load-shedding relays

PURPOSE

Metrology-grade three-phase measurement on an ATM90E32AS, with an alarm engine that can shed load locally. Closer to switchboard protection that happens to count kWh than to a plain energy meter.



SPECIFICATIONS

Measurement	3-phase + neutral: Urms, Irms, signed P/Q, S, PF, angle, frequency, THD
Energy	Primary-side Wh / varh, separate import and export
Current inputs	Current-output CTs, ≤60 mA RMS secondary — not 1 A or 5 A types
Full scale	120 / 60 / 30 mA at PGA ×1 / ×2 / ×4 (default ×2)
Relays	2 × SPDT, 3 A @ 250 VAC — None / Modbus / Alarm modes
Isolation	5 kV RMS between MCU and measurement side
Controls	4 buttons, 9 LEDs
Bus	RS-485 Modbus RTU, default ID 3 @ 19200

WHAT MAKES IT DIFFERENT

Power quality, not just consumption

THD and harmonics per phase, sag, overvoltage, phase loss and phase reversal detection. The faults that kill equipment, visible before they do.

Load shedding without the controller

The on-board alarm engine drives the relays on threshold. It sheds load whether or not the controller or the server is reachable.

Per-phase alarm engine

L1–L3 and total, each with Alarm / Warning / Event levels and acknowledge.

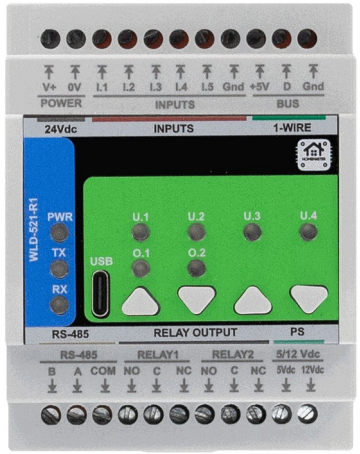
WATER

WLD-521-R1

Leak detection with valve shut-off, water metering and heat metering

PURPOSE

Multi-zone leak detection that closes the valve by itself, plus pulse water metering and indicative heat-energy metering from flow and temperature difference. Safety logic runs autonomously on the module.



SPECIFICATIONS

Inputs	5 × opto-isolated, dry contact or pulse flowmeter
Relays	2 × SPDT, 3 A @ 250 VAC — valve control
Water metering	Pulse counting from a standard meter
Heat metering	Flow × ΔT from 1-Wire supply and return sensors; instantaneous W and accumulated kWh
1-Wire	Supply and return temperature sensors
Bus	RS-485 Modbus RTU

WHAT MAKES IT DIFFERENT

The valve closes with the controller unplugged

Detection and shut-off are entirely local. Nobody home, internet down, router off — it still closes. This is the argument that sells the system.

Heat metering on a general-purpose module

Instantaneous power and accumulated kWh for underfloor or radiator loops. Normally a separate expensive instrument. Indicative, not MID-certified for billing.

Multi-zone

Five independent zones — kitchen, bathroom, boiler, utility, technical room — on one module.

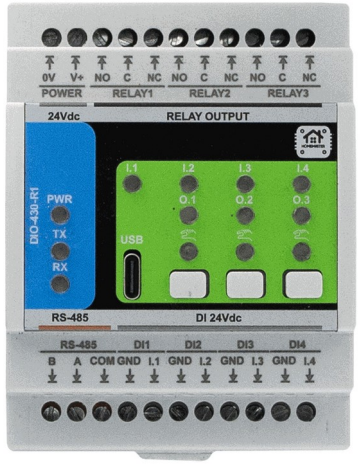
I/O

DIO-430-R1

Digital I/O module with standalone local logic

PURPOSE

The workhorse: wall switches drive relays directly through the module, with or without a controller. Gesture counters, interlock and staircase timers run on board.



SPECIFICATIONS

Digital inputs	4 × 24 V, IEC 61131-2 (ISO1212), module-wetted dry contact
Relays	3 × SPDT, 3 A @ 250 VAC — use contactors above 3 A
Input logic	Per-input Maintained / Momentary; short, long, double press actions
Buttons	3 (2 user-configurable)
LEDs	13 LEDs (PWR, 3 user, RX, TX, 3 relay, 4 DI)
Protection	PTC fuse, TVS surge, reverse polarity; opto-isolated relay drivers
Extras	All-Off, child lock, auto-off staircase timer, shutter interlock
Bus	RS-485 Modbus RTU

WHAT MAKES IT DIFFERENT

Wall switches keep working, always

Input to relay mapping runs on the module. A reboot, an update or a cut cable upstream does not turn the house into a house without light switches.

Last command wins, no mode to flip

Wired input, front button and Home Assistant all control the same relay simultaneously. There is no local-versus-remote switch to get wrong.

Shutter interlock on board

Two outputs can never be energised together — the motor is protected by hardware logic, not by an automation someone might delete.

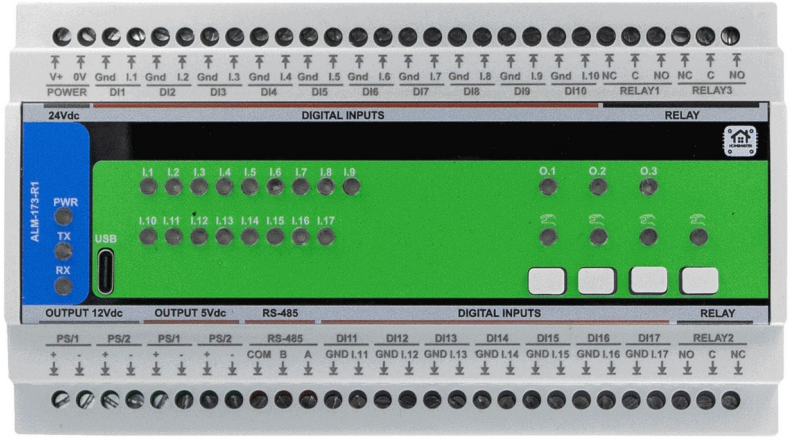
SECURITY

ALM-173-R1

17-zone sensor hub with on-board alarm engine and detector power

PURPOSE

Connects seventeen wired sensors to Home Assistant, a PLC or any SCADA from one module — and powers them itself from built-in isolated 12 V and 5 V rails. Not a certified intruder alarm; an automation and monitoring module.



SPECIFICATIONS

Inputs	17 × opto-isolated, 5 V DC signalling, 5300 VRMS isolation per channel
Relays	3 × SPDT, 3 A @ 250 VAC
Detector power	Isolated +12 V and +5 V rails on the module
Alarm groups	3 groups plus Any-Alarm aggregate; latched or non-latched
Controls	4 buttons, 27 LEDs; acknowledge and relay override
Extras	Local arming, bell cut-off, per-zone alarm memory
Bus	RS-485 Modbus RTU

WHAT MAKES IT DIFFERENT

Sensor power is on the module

Wired PIRs, glass-break and smoke detectors need power as well as a contact. Normally that means a second PSU and another distribution block. Here both rails are on board — cabinet space and one less thing to buy.

The alarm engine runs offline

Zones, groups, latching and relay actions execute on the module. The server going down does not disarm the house.

Seventeen zones on one DIN module

Where a typical panel needs an expander, this is one unit of cabinet width.

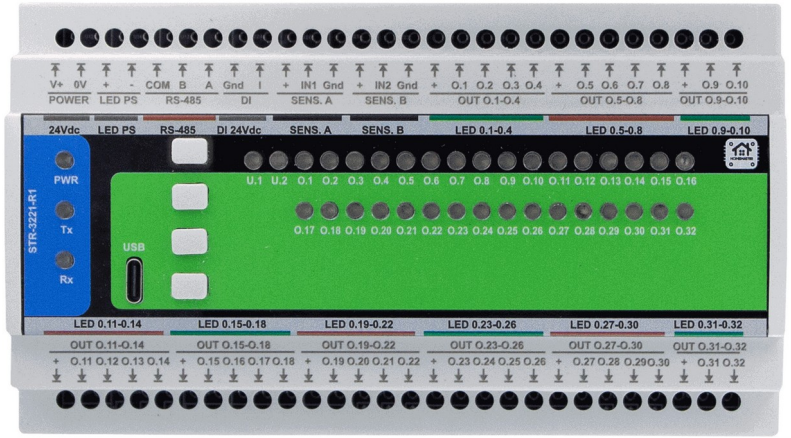
LIGHTING

STR-3221-R1

32-channel LED controller for staircase and architectural lighting

PURPOSE

A high-density lighting node. Channels and inputs are published on Modbus; the controller (MiniPLC / MicroPLC or Home Assistant) sets the sequence.



SPECIFICATIONS

Outputs	32 × low-side MOSFET, 12–24 V, nine groups with (4+4+2 / 4+4+4+4+4+4+2); 1.5 A/ch, 18 A separate + rails
Drivers	PWM
Presence inputs	2 × presence inputs (SFH6156, module-supplied)
Digital input	1 × 24 V, IEC 61131-2 (ISO1212)
Buttons	4; state published on Modbus
Control	Sequence is set by the controller; channels and inputs are published on Modbus
Bus	RS-485 Modbus RTU

WHAT MAKES IT DIFFERENT

Thirty-two independently dimmable channels

Chinese stair controllers exist, but they are standalone boxes with no smart-home integration at all. This one sits on the bus and appears in Home Assistant like any other module.

Thirty-two channels on one Modbus address

Channels and inputs are published on Modbus. The controller (MiniPLC / MicroPLC or Home Assistant) sets the sequence.

Not only for stairs

Thirty-two general-purpose 12/24 V channels at 1.5 A also drive underfloor heating valve actuators, cabinet lighting or garden circuits.

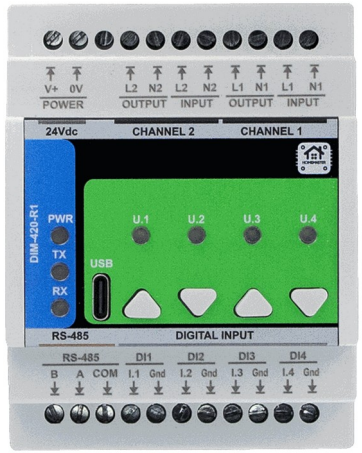
LIGHTING

DIM-420-R1

Two-channel phase-cut AC dimmer with four inputs

PURPOSE

Mains dimming with leading or trailing edge selectable per channel, plus four digital inputs with full press logic. Local dimming keeps working offline.



SPECIFICATIONS

Dimming outputs	2 × MOSFET phase-cut AC, leading or trailing per channel
Thresholds	Lower and upper limits per channel, zero-cross sync
Digital inputs	4 × 24 V, IEC 61131-2 (ISO1212)
Press types	Short / Long / Double / Short-then-Long
Protection	PTC fuse, TVS surge, reverse polarity
Bus	RS-485 Modbus RTU
Limitation	No per-channel power or energy metering

WHAT MAKES IT DIFFERENT

Leading or trailing per channel

Set the dimming mode to match the load — one channel for a transformer, another for LED drivers, on the same module.

Press logic on the module

Short press toggles, long press dims, double press goes to full. It works the same whether the server is running or not.

LIGHTING

RGB-621-R1

RGB and tunable white LED controller with scene engine

PURPOSE

Five PWM channels for RGB plus tunable white, with 12-bit gamma-corrected dimming and an on-module scene and gesture engine.



SPECIFICATIONS

PWM outputs	5 × N-channel MOSFET, 12 or 24 V — R / G / B / CW / WW
Dimming	12-bit with gamma correction
Relay	1 × SPST-NO, 3 A @ 250 VAC
Digital inputs	2 × 24 V, IEC 61131-2 (ISO1212)
Buttons	2 for local control
LEDs	8 indicators
Bus	RS-485 Modbus RTU

WHAT MAKES IT DIFFERENT

- **Tunable white, not just colour**

  Cold and warm white channels mean the colour temperature follows the time of day — the part of lighting design clients actually notice.

- **Scenes stored on the module**

  The gesture engine and scenes run locally. A wall button still cycles scenes with the network down.

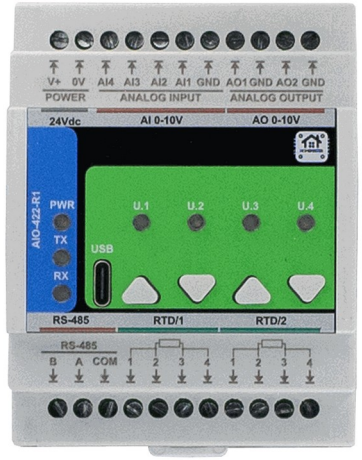
ANALOGUE

AIO-422-R1

Analogue I/O with RTD inputs

PURPOSE

Analogue sensing and actuation in one DIN module — 0–10 V in and out plus two precision RTD channels, without stacking separate boards.



SPECIFICATIONS

Analogue inputs	4 × 0–10 V, 16-bit (ADS1115) — voltage only, no 4–20 mA
Analogue outputs	2 × 0–10 V, 12-bit (MCP4725)
RTD	2 × MAX31865, PT100 or PT1000, 2/3/4-wire
Buttons	4 with configurable actions
LEDs	plus PWR, RX, TX
Bus	4 user LEDs plus PWR, RX, TX

WHAT MAKES IT DIFFERENT

RTD and 0–10 V on the same module

Read a buffer tank with PT1000 and drive a modulating valve with 0–10 V from one unit of cabinet width.

Measurement stays on the module

AI and RTD values are always measured locally; a controller only reads them. Polling gaps do not create holes in the data.

HEATING

OpenTherm Gateway

OpenTherm master with full boiler telemetry

PURPOSE

An ESP32 OpenTherm master — not a Modbus slave — exposing full boiler telemetry in Home Assistant and driving modulating, weather-compensated heating.



SPECIFICATIONS

Protocol	OpenTherm master (OT+ / OT−)
Telemetry	Modulation level, flame state, flow and return temperatures, DHW, fault codes
Control	Modulating setpoint, weather compensation
1-Wire	2 buses for room and outdoor sensors
Relay	1 × C/NC only (SPST-NC), 3 A @ 250 VAC
Network	Wi-Fi with Improv onboarding
Power	24 V DC nominal or 85–265 V AC

WHAT MAKES IT DIFFERENT

Modulation instead of on-off

The boiler runs at the output the house actually needs rather than cycling. Less gas, longer boiler life, and no temperature swing.

The fault codes reach the owner

Boiler errors appear in Home Assistant as they happen, not when someone notices the house is cold.

Weather compensation without the manufacturer's controller

Outdoor sensor on 1-Wire, curve in the gateway. No proprietary thermostat, no subscription.

CONTROLLER

MicroPLC

Compact ESP32 Modbus RTU master

PURPOSE

The entry-point controller. Same ESP32 and same firmware as the MiniPLC — the difference is onboard I/O, mains input and Ethernet, not processing power. Two uses: drive a bus of expansion modules, or talk directly to third-party Modbus RTU equipment such as heat recovery units, inverters and heat pumps.



SPECIFICATIONS

Relay	1 × C/NC only (SPST-NC), 3 A @ 250 VAC
Digital input	1 × 24 V
1-Wire	1 bus
Extras	RTC
Network	Wi-Fi with Improv onboarding
Bus	RS-485 Modbus RTU master
Firmware	ESPHome pre-installed, identical to MiniPLC

WHAT MAKES IT DIFFERENT

A route into third-party Modbus equipment

More and more HRV units, inverters and heat pumps ship with a Modbus port and no sensible way into Home Assistant. This is a clean answer to that.

Same firmware as the flagship

Configurations, packages and know-how transfer directly between MicroPLC and MiniPLC. Nothing to relearn.

COMMERCIAL

Terms and compliance

Partner and distributor conditions

PARTNER TERMS

Discount	20% off net from the first order
Minimum order	None
Annual quota	None
Demo kit	One controller plus up to three modules at 40% off net
Payment	Bank transfer against invoice, first order prepaid
Paperwork	Invoice only, no contract
Warranty	24 months on manufacturing defects
RMA	Replacement and return shipping at our cost; batch returns
Delivery	1–3 days within the EU from Ploiești
Stock items	Shipped within two working days
Not in stock	Within two weeks

COMPLIANCE

CE marking	Signed declaration of conformity per module
EMC	EN 55032 Class B emission, EN 55035 immunity
Laboratory	Idvorsky Laboratories, Belgrade — accredited
Electrical safety	EN 62368-1
Radio (controllers)	RED via Kiwa Nederland, Notified Body 0063
Cybersecurity	EN 18031-1:2024, RED Article 3(3)(d)
RoHS	Technical file per EN IEC 63000
Not certified	No UL, no FCC — EU market
Switzerland	BAKOM accepts CE, no recertification
Turkey	CE accepted, A.TR at customs

WORTH KNOWING BEFORE YOU SPECIFY

- RS-485 is transient-protected (TVS, PTC, common-mode choke) but not galvanically isolated — keep the bus inside one cabinet and one earthing system; use an external isolator where it crosses into another installation.
- Relay contacts are rated 3 A at 250 VAC per channel. Use interposing contactors above that.
- Operating range 0 to +40 °C — heated indoor cabinets.
- Module firmware is flashed over USB-C at the cabinet; controllers update over the air through ESPHome. Module configuration cannot yet be exported.
- 7–10 modules per controller keeps polling fast enough that a button press feels instant. Scale with more controllers, not a longer bus.